IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY
COLLEGE OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
CSC 305 – Software Engineering | Semester 1 – 2026/2027 | Section 5MA1 | Group 3

Software Requirements Specification (SRS)

InternSync – Cooperative Training Coordination and Tracking System

Project 4 · Version 1.0 · October 2026

Team Information

Team Member	Student ID	Project Role
Salam Alsaour (Team Leader)	2250009168	Team leader; backend and AI services developer
Khalid Alajmi	2250004309	Frontend and backend developer
Ali Alghafly	2250002460	Frontend developer
Ahmed Kanaan	2250009170	AI services developer; deployment lead
Abdullah Abalhareth	2250005266	QA and documentation lead
Rakan Alzahrani	2250000631	UI/UX designer

Revision History

Version	Date	Description	Author
1.0	Oct 10, 2026	First baseline submitted as Project 4 (SRS).	Group 3

Table of Contents

1. Introduction....4
1.1 Purpose of this Document....4
1.2 System Purpose....4
1.3 Scope....4
1.4 Definitions, Acronyms, and Abbreviations....5
1.5 References....6
1.6 Requirement Conventions....6
2. Stakeholders and User Classes....7
2.1 Stakeholders....7
2.2 User Classes and Characteristics....7
2.3 Access Rights by Role....8
3. Overall Description....9
3.1 Product Perspective....9
3.2 Co-op Life Cycle Supported....9
3.3 Use Case Overview....10
3.4 Operating Environment....10
3.5 User Documentation....10
4. Functional Requirements....11
4.1 Accounts, Authentication, and Access Control....11
4.2 Training Cycles and Configuration....11
4.3 Student Profile, CV, and Data Import....12
4.4 Vacancies and Placement....12
4.5 Matching and Explanation....13
4.6 Weekly and Final Reports....14
4.7 Attendance....15
4.8 Evaluations....15
4.9 Approval Workflow and Grades....16
4.10 Similarity Detection....17
4.11 Notifications....17
4.12 Dashboards and Statistics....18
4.13 Exports....18
4.14 Completion Certificates....19
4.15 Language and PWA Features....19
4.16 Audit Trail....20
5. Non-Functional Requirements....20
5.1 Performance....20
5.2 Security....20
5.3 Privacy and Data Protection....21
5.4 Usability and Accessibility....21
5.5 Localization....22
5.6 Reliability and Availability....22
5.7 Maintainability and Portability....22
5.8 AI Quality and Transparency....23
6. External Interface Requirements....23
6.1 User Interfaces....23
6.2 Hardware Interfaces....24
6.3 Software Interfaces....24

6.4 Communication Interfaces..... 25
7. Constraints.....26
8. Assumptions and Dependencies.....26
8.1 Assumptions..... 26
8.2 Dependencies..... 27
9. Requirement Prioritization..... 27
9.1 Method..... 27
9.2 Priority Summary..... 28
9.3 Mapping to Increments..... 28
9.4 Descoping Order..... 29
10. Requirement Validation Considerations.....29
10.1 Validation and Verification..... 29
10.2 Validation Techniques..... 29
10.3 Requirement Quality Checklist..... 30
10.4 Verification Methods..... 30
10.5 Validation Risks and Open Issues..... 30
10.6 Baseline and Change Control..... 31
Appendix A. Traceability Matrix.....31

1. Introduction

1.1 Purpose of this Document

This Software Requirements Specification (SRS) defines what InternSync must do and the qualities it must have. It turns the functions (F1–F19) and objectives (O1–O10) approved in the Project Proposal into numbered, testable requirements. Once approved, it is the baseline for design, implementation, and testing, and the reference for the change control described in the Software Project Management Plan (SPMP).

The intended readers are the course supervisor, who evaluates the specification; the development team, who design and build from it; the QA lead, who derives test cases from it; and the stakeholder representatives who take part in validating it (Section 10).

1.2 System Purpose

Cooperative training (Co-op) and summer internships are coordinated today through email, spreadsheets, and paper or PDF forms. Vacancies are circulated informally, weekly reports arrive as attachments, company evaluations are filled in once at the end, and grade sheets are compiled by hand (problems P1–P6 in the Proposal). The result is no single, reliable record of what happened during a training cycle.

InternSync is a role-based web platform, delivered as a responsive Progressive Web App (PWA) in Arabic and English, that manages the whole co-op life cycle in one place: publishing vacancies and matching students to them, weekly follow-up, standard evaluation, multi-stage grade approval, and official exports. Its purpose is to replace scattered manual work with one auditable record from which the official outputs are generated directly. The AI features (matching, skill gaps, similarity detection) support decisions; people make every final decision.

1.3 Scope

This release covers functions F1–F19 of the Proposal. Table 1.1 lists them with their priority and the section of this SRS that specifies them.

Table 1.1 – Functions in scope

ID	Function	Priority	SRS section
F1	Role-based accounts, authentication, and authorization for four roles	Must	4.1
F2	Student profile and CV upload with skill extraction; CSV import of student, course, and staff data	Must	4.3
F3	Vacancy catalogue, applications, and placement confirmation	Must	4.4
F4	Matching engine that ranks vacancies against the student profile	Must	4.5
F5	Weekly and final report submission with deadlines and revision requests	Must	4.6
F6	Weekly attendance confirmation by the company supervisor	Must	4.7
F7	Periodic and final evaluations using one standard rubric	Must	4.8
F8	Multi-stage approval workflow for reports and grades	Must	4.9
F9	Role-specific dashboards with pending actions and status	Must	4.12
F10	Arabic/English interface with RTL and LTR support	Must	4.15
F11	Match explanation, skill-gap analysis, and development plan	Should	4.5
F12	Similarity detection between reports stored in the system	Should	4.10

ID	Function	Priority	SRS section
F13	Scheduled notifications: reminders, late flags, escalation	Should	4.11
F14	Export of grade sheets and summary reports (PDF and Excel)	Should	4.13
F15	PWA features: installable app, web push, cached app shell	Should	4.15
F16	Digital completion certificate with QR code and public verification	Could	4.14
F17	Mutual evaluation: the student evaluates the training entity	Could	4.8
F18	End-of-term survey and placement-to-job-offer statistics	Could	4.12
F19	Evaluation-leniency alert for companies with at least 5 evaluations	Could	4.12

Supporting requirements that serve several functions — training-cycle configuration (Section 4.2) and the audit trail (Section 4.16) — are also in scope.

Out of scope for this release (the “Won’t” category in Section 9):

- Live integration with the university student information system; student and staff data are imported from CSV files.
- Writing grades into the university system; grades leave InternSync as exported files only.
- A native mobile application; the responsive PWA covers phone use.
- Contracts, payroll, insurance, or any legal or HR processing with companies.
- Plagiarism detection against the internet or external databases, and detection of AI-generated text.
- Biometric or GPS-based attendance verification.
- Real-time chat; communication is limited to notifications and comments on submissions.
- Statistical calibration of grades across companies; a standard rubric is used instead.

1.4 Definitions, Acronyms, and Abbreviations

Table 1.2 – Definitions

Term	Definition
Co-op	Cooperative training: a supervised training period at a company that is part of the degree plan.
Training cycle	One term’s run of co-op or summer training, with its own dates, deadlines, rubric, and grade weights.
Placement	Confirmed assignment of a student to a vacancy, a company supervisor, and an academic supervisor.
Company supervisor (CS)	Company employee who supervises the student during training.
Academic supervisor (AS)	Faculty member who follows the student’s progress and proposes the grade.
Coordinator (CO)	Staff member of the college co-op unit; also performs system administration in this release.
Student (ST)	Student enrolled in a training cycle.
Rubric	Fixed set of evaluation criteria, each rated on a 1–5 scale with a description for every level.
Skill vocabulary	List of normalized skills with English and Arabic names and synonyms, maintained by the coordinator.

Term	Definition
Embedding	Numeric vector representation of text, used to compare meaning (CV with vacancy, report with report).
Fit score	Value from 0 to 100 used to rank vacancies for a student.
Similarity score	Percentage of a report's text that overlaps with another stored report.
Default	Value set by this SRS that the coordinator can change without code changes (NFR-MNT-03).
MoSCoW	Prioritization into Must, Should, Could, and Won't.
PWA	Progressive Web App: a web application that can be installed and can work partly offline.
RTL / LTR	Right-to-left (Arabic) / left-to-right (English) layout.
RBAC	Role-based access control.
JWT	JSON Web Token, used to authenticate API requests.
PDPL	Saudi Personal Data Protection Law.
UAT	User acceptance testing.
FR / NFR	Functional requirement / non-functional requirement.
SPMP	Software Project Management Plan.

1.5 References

1. InternSync – Project Proposal, CSC 305 Group 3, Oct 2, 2026.
2. InternSync – Software Project Management Plan (SPMP), CSC 305 Group 3, Oct 3, 2026.
3. InternSync – Project Idea Form, CSC 305 Group 3, Sep 5, 2026 (superseded by the Proposal where they differ, e.g., four roles instead of five).
4. ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.
5. Personal Data Protection Law (PDPL), Kingdom of Saudi Arabia, and its implementing regulations.
6. W3C, Web Content Accessibility Guidelines (WCAG) 2.1.

1.6 Requirement Conventions

- Every requirement has a unique ID: **FR-<area>-<nn>** for functional and **NFR-<category>-<nn>** for non-functional requirements. IDs are never reused; a removed requirement keeps its ID and is marked "Deleted".
- Each requirement states one behavior using "shall" and has an acceptance criterion from which test cases are derived (Section 10).
- Priority is Must, Should, or Could, as defined in Section 9.
- Each group of requirements names its source function (F) and objective (O) from the Proposal; Appendix A gives the full traceability.
- Role abbreviations: ST student, CS company supervisor, AS academic supervisor, CO coordinator.

2. Stakeholders and User Classes

2.1 Stakeholders

Table 2.1 lists the people and groups affected by InternSync, what they need, and how their needs shape the requirements.

Table 2.1 – Stakeholders

Stakeholder	Type	Needs and interests	Interaction with the system	Effect on requirements
Student	Primary user	Find suitable vacancies and know which skills are missing; submit reports easily; get feedback during training; see grade and certificate.	Profile and CV, applications, weekly and final reports, evaluation of the company, dashboard.	Mobile-friendly, simple flows (NFR-USA); advisory, explained matching.
Company training supervisor	Primary user (external, occasional)	Spend little time: confirm attendance and evaluate quickly; reach qualified students.	Vacancies, applicant review, weekly attendance, periodic and final evaluation.	One-action attendance (FR-ATT-01); email reminders; nothing to install.
Academic supervisor	Primary user (faculty)	Review reports efficiently, notice copied work, propose grades with a clear record.	Report review, revision requests, similarity flags, grade entry.	Approval workflow (4.9); similarity is advisory (FR-SIM-03).
Co-op coordinator (co-op unit)	Primary user; system administrator in this release	Run cycles without manual work; control deadlines; produce official outputs; handle escalations.	Accounts and CSV import, cycles, rubric, vacancies, placements, grade approval, exports, audit log.	Configuration (4.2), exports (4.13), audit trail (4.16).
College administration	Indirect	Accurate program statistics and placement outcomes.	Receives term summaries exported by the coordinator; no account in this release.	FR-EXP-02, FR-DSH-03.
Partner companies and employers	Indirect	A reliable channel to students; verify that a student completed training.	Through company supervisor accounts; public certificate verification.	FR-CER-01 – 03.
CSC 305 course supervisor	Project stakeholder	A complete, testable SRS and delivery on schedule.	Reviews deliverables at each milestone.	Constraints (7) and prioritization (9).
Development team (Group 3)	Project stakeholder	A scope that fits about 480 person-hours.	Designs, builds, and tests the system.	Constraints (7) and descoping order (9.4).

2.2 User Classes and Characteristics

The expected numbers describe one training cycle in the demonstration environment and are used as the sizing basis for the performance requirements (Section 5.1).

Table 2.2 – User classes

User class	Expected number per cycle	Frequency of use	Technical skill	Main device
Student	Up to 300	Daily while searching; weekly during training	Medium to high	Phone and laptop

User class	Expected number per cycle	Frequency of use	Technical skill	Main device
Company supervisor	Up to 100	Weekly, a few minutes	Varies; may be non-technical	Laptop; phone through email links
Academic supervisor	Up to 30	Weekly	Medium	Laptop
Coordinator	1 to 3	Daily during setup and grading	Medium	Laptop
Public verifier	Not known	Rare	Any	Phone (QR scan)

2.3 Access Rights by Role

Table 2.3 defines which role may perform each action. “Own” means the user’s own records; “Assigned” means the records of students placed under that supervisor. FR-AUT-05 requires the server to enforce this table.

Table 2.3 – Permission matrix

Action	ST	CS	AS	CO	Public
Manage accounts; import CSV files	—	—	—	All	—
Configure cycles, rubric, grade weights, skill vocabulary	—	—	—	All	—
Edit profile, upload CV, confirm skills	Own	—	—	—	—
Create a vacancy	—	Own company	—	All	—
Publish, return, or close a vacancy	—	—	—	All	—
View ranked vacancies and apply	Own	—	—	—	—
Review applicants of a vacancy	—	Own vacancies	—	All	—
Confirm or cancel a placement	—	—	—	All	—
Submit weekly and final reports	Own	—	—	—	—
View reports	Own	Assigned (read)	Assigned	All	—
Accept a report or request revision	—	—	Assigned	—	—
Record attendance	—	Assigned	—	Unlock only	—
Submit periodic and final evaluations	—	Assigned	—	Reopen only	—
View similarity flags and record a decision	—	—	Assigned	View all	—
Enter and submit a grade	—	—	Assigned	—	—
Approve, return, or reopen a grade	—	—	—	All	—
View a final grade	Own	—	Assigned	All	—
Evaluate the training entity	Own	—	—	View all	—
Export grade sheets and reports	—	—	Assigned (records)	All	—
Issue or revoke a certificate	—	—	—	All	—
Verify a certificate	—	—	—	—	Yes

Action	ST	CS	AS	CO	Public
View the audit log	—	—	—	All	—

3. Overall Description

3.1 Product Perspective

InternSync is a new, self-contained system that replaces the current mix of email, spreadsheets, and paper forms. It has no live connection to university systems: student and staff data enter through CSV files, and grades leave as exported PDF and Excel files. Figure 3.1 shows the system boundary, its users, and the external services it depends on.

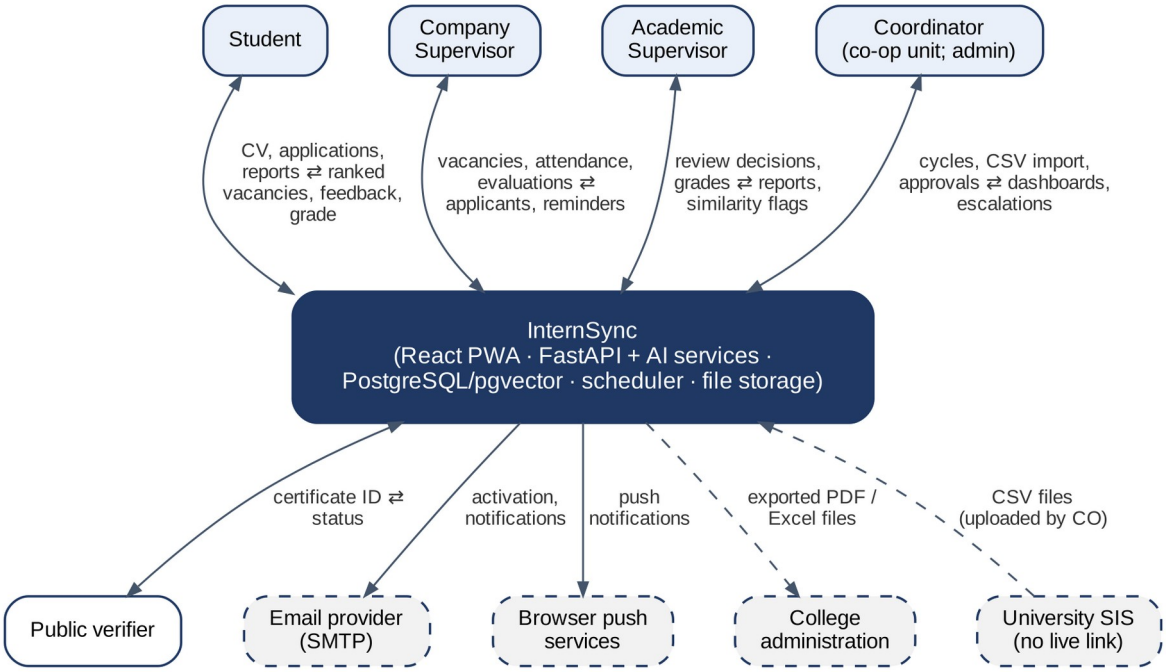


Figure 3.1 – Context diagram

Inside the boundary, a React PWA client talks to a single Python/FastAPI backend that also hosts the AI services (skill extraction, matching, similarity). PostgreSQL with pgvector stores relational data and embeddings, a scheduler runs reminder and escalation jobs, and file storage keeps CVs and attachments. These components are named only to show the boundary; their design is described in the design document (task T2).

3.2 Co-op Life Cycle Supported

1. The coordinator opens a training cycle, imports students and academic supervisors from CSV files, and creates company supervisor accounts.
2. Company supervisors submit vacancies; the coordinator reviews and publishes them.
3. Students build their profiles, upload CVs, confirm the extracted skills, view vacancies ranked by fit with an explanation, and apply.
4. Company supervisors accept or decline applicants; the coordinator confirms placements and assigns academic supervisors.

5. During training, students submit weekly reports, company supervisors record attendance weekly and submit the periodic evaluation, and the system sends reminders and late flags.
6. Academic supervisors review reports with the similarity indicator and accept them or request revisions.
7. At the end of training, the student submits the final report, the company supervisor submits the final evaluation, and the grade passes through the approval workflow.
8. The coordinator exports grade sheets and summaries and issues certificates; students evaluate the training entity and answer the outcome survey.

3.3 Use Case Overview

Table 3.1 – Use cases

UC	Use case	Primary actor	Requirements
UC-01	Open a training cycle and import users	CO	FR-CYC-01 – 04, FR-PRF-01, FR-AUT-02
UC-02	Activate account, log in, reset password	All roles	FR-AUT-01 – 04
UC-03	Build profile and confirm CV skills	ST	FR-PRF-02 – 05
UC-04	Submit and publish a vacancy	CS, CO	FR-VAC-01, FR-VAC-02
UC-05	View ranked vacancies and apply	ST	FR-VAC-03 – 05, FR-MAT-01 – 07
UC-06	Review applicants and confirm placement	CS, CO	FR-VAC-06 – 08
UC-07	Submit a weekly or final report	ST	FR-REP-01 – 04
UC-08	Review a report and check similarity	AS	FR-REP-05 – 07, FR-SIM-01 – 03
UC-09	Record weekly attendance	CS	FR-ATT-01 – 04
UC-10	Submit a periodic or final evaluation	CS	FR-EVL-01 – 04
UC-11	Enter, submit, and approve a grade	AS, CO	FR-WFL-02 – 07
UC-12	Export grade sheets and summaries	CO	FR-EXP-01 – 03
UC-13	Issue and verify a certificate	CO, Public	FR-CER-01 – 03
UC-14	Evaluate the training entity and answer the survey	ST	FR-EVL-05, FR-DSH-03

3.4 Operating Environment

- **Client:** current versions of Chrome, Edge, Firefox, and Safari on desktop, and Chrome and Safari on phones (NFR-PRT-01); installable as a PWA.
- **Server:** one Linux virtual machine running Docker; reference demonstration server with 2 vCPU, 4 GB RAM, and 40 GB disk. No GPU is required.
- **Software:** Python with FastAPI; React with Vite; PostgreSQL 16 with pgvector; an open-source multilingual embedding model (for example, multilingual-e5-small) running on the server.
- **Time zone:** all deadlines are defined and evaluated in Saudi Arabia time (Asia/Riyadh, UTC+3).

3.5 User Documentation

- A bilingual quick-start guide for each role (PDF, at most 4 pages each).
- Help text on the main forms (report, attendance, evaluation, grade).
- A deployment and setup guide for the technical team (NFR-PRT-02).

4. Functional Requirements

Functional requirements are grouped by area. In each table, the acceptance criterion states the observable result that a test must confirm. Values marked “default” are configurable (NFR-MNT-03).

4.1 Accounts, Authentication, and Access Control

Source: F1 · Objective O1

ID	Requirement	Priority	Acceptance criterion
FR-AUT-01	The system shall support four roles — Student, Company Supervisor, Academic Supervisor, and Coordinator — and each account shall hold exactly one role. The Coordinator also performs system administration in this release.	Must	A user of each role logs in and sees only the menu items of that role.
FR-AUT-02	The system shall not offer public self-registration. The Coordinator shall create accounts one at a time or through CSV import (FR-PRF-01), and the system shall email each new user a one-time activation link valid for 72 hours (default).	Must	No registration page exists; an activation link works once; a used or expired link is rejected with a message.
FR-AUT-03	Users shall log in with email and password. A session shall end after 30 minutes of inactivity (default), and users shall be able to log out at any time.	Must	After 30 idle minutes the next action returns to the login page; after logout the old token is rejected.
FR-AUT-04	The system shall let a user reset a forgotten password through an emailed one-time link valid for 60 minutes.	Must	The link resets the password once; the old password stops working; an expired link is rejected.
FR-AUT-05	The system shall enforce the permission matrix of Table 2.3 on the server for every request: supervisors reach only the records of students assigned to them, and students only their own records.	Must	Automated permission tests cover every API endpoint for every role, with 0 failing tests (O1).
FR-AUT-06	The Coordinator shall be able to deactivate and reactivate accounts. A deactivated user cannot log in, and the records that user created remain unchanged.	Must	Login of a deactivated user fails; the user’s past reports and evaluations stay visible to authorized users.

4.2 Training Cycles and Configuration

Source: supports F3, F5, F7, F8 · Objectives O3, O4

ID	Requirement	Priority	Acceptance criterion
FR-CYC-01	The Coordinator shall create a training cycle with a name, type (Co-op or Summer), start and end dates, number of training weeks, weekly report deadline (weekday and time), and due dates for the periodic evaluation, final evaluation, and final report.	Must	Creating a 12-week cycle generates 12 weekly deadlines on the configured weekday and time.
FR-CYC-02	The Coordinator shall define the grade components of a cycle and their weights; the system shall reject weights that do not sum to 100%.	Must	Weights 40/30/20 are rejected; 40/30/30 are saved.

ID	Requirement	Priority	Acceptance criterion
FR-CYC-03	The Coordinator shall define one evaluation rubric per cycle: a list of criteria, each with a description for every level of a 1–5 scale. After the first evaluation of the cycle is submitted, the rubric shall become read-only.	Must	After one evaluation is submitted, editing the rubric is refused.
FR-CYC-04	The Coordinator shall maintain the skill vocabulary: each skill has an English name, an Arabic name, synonyms, and optional learning resources (title, link, estimated effort).	Must	A skill with the synonym “JS” is recognized in a CV containing “JS” and is shown as “JavaScript”.
FR-CYC-05	The Coordinator shall be able to close a cycle, which makes its records read-only; reopening a closed cycle requires a written reason, recorded in the audit log.	Should	Edits in a closed cycle are refused; reopening without a reason is refused.

4.3 Student Profile, CV, and Data Import

Source: F2 · Objectives O2, O7

ID	Requirement	Priority	Acceptance criterion
FR-PRF-01	The Coordinator shall import students and academic supervisors from CSV files in the formats of Section 6.3.1. The system shall validate every row, create accounts for valid rows, and show an import report listing each rejected row with its reason (missing field, invalid email, duplicate ID).	Must	A 50-row file with 3 invalid rows creates 47 accounts and lists exactly the 3 rejected rows with reasons.
FR-PRF-02	A student shall maintain a profile with major and GPA (imported and read-only), confirmed skills, languages, and areas of interest.	Must	The student can edit skills and interests but not major or GPA.
FR-PRF-03	A student shall upload a CV as a PDF or DOCX file of up to 5 MB. The latest CV is used for matching, and earlier versions are kept.	Must	A 6 MB file or another file type is refused; after a second upload both versions appear in the CV history.
FR-PRF-04	After each CV upload, the system shall extract candidate skills from the CV text (Arabic or English), normalize them with the skill vocabulary, and show them to the student, who can add, edit, or remove skills before confirming. Only confirmed skills shall be used for matching.	Must	On a labelled set of 20 CVs (10 Arabic, 10 English), at least 70% of the skills listed by the annotators appear in the extracted list.
FR-PRF-05	If no text can be extracted from a CV (for example, a scanned image), the system shall inform the student and allow manual skill entry.	Must	Uploading an image-only PDF shows the message and an empty, editable skill list.

4.4 Vacancies and Placement

Source: F3 · Objective O1

ID	Requirement	Priority	Acceptance criterion
FR-VAC-01	A Company Supervisor or the Coordinator shall create a vacancy with title, company, city, description, required skills, preferred skills, eligible majors, optional minimum GPA, number of seats, application deadline, and cycle.	Must	A vacancy missing a required field cannot be saved; a saved vacancy shows all fields.

ID	Requirement	Priority	Acceptance criterion
FR-VAC-02	A vacancy has one of the statuses Draft, Pending Review, Published, or Closed. A vacancy created by a Company Supervisor shall become visible to students only after the Coordinator publishes it; the Coordinator may return it with a comment. A vacancy created by the Coordinator is published directly.	Must	Students cannot see a Pending Review vacancy; after publication it appears in their list.
FR-VAC-03	Students shall browse the published vacancies of their cycle, search by keyword, and filter by city, company, eligible major, and skill.	Must	Each filter returns only vacancies that satisfy it (checked on seeded data).
FR-VAC-04	A student shall apply to an open vacancy for which they are eligible, before its deadline. The system shall prevent duplicate applications and limit active applications to 5 per student (default).	Must	A second application to the same vacancy, an application after the deadline, and a sixth active application are each refused with a message.
FR-VAC-05	A student shall be able to withdraw an application until a decision has been made on it.	Should	The withdrawn application leaves the applicant list and frees an application slot.
FR-VAC-06	A Company Supervisor shall view the applicants to their own vacancies (profile, confirmed skills, CV) and mark each applicant Accepted or Declined.	Must	The supervisor sees applicants of own vacancies only; the decision is visible to the student and the Coordinator.
FR-VAC-07	The Coordinator shall confirm the placement of an accepted applicant by assigning the student to the vacancy, its Company Supervisor, and an Academic Supervisor. A student shall have at most one confirmed placement per cycle; confirming it withdraws the student's other active applications, and a vacancy closes automatically when its seats are filled.	Must	A second placement for the same student is refused; filling the last seat changes the vacancy to Closed.
FR-VAC-08	The Coordinator shall be able to change a student's Academic Supervisor or cancel a placement, giving a mandatory reason.	Should	The action without a reason is refused; the change appears in the audit log.

4.5 Matching and Explanation

Source: F4, F11 · Objectives O2, O7

ID	Requirement	Priority	Acceptance criterion
FR-MAT-01	For a student with confirmed skills, the system shall rank the open vacancies of the student's cycle for which the student is eligible by a fit score from 0 to 100, in descending order.	Must	On a labelled set of at least 20 student profiles, at least one relevant vacancy is in the top 5 for at least 80% of profiles (O2).
FR-MAT-02	The fit score shall combine the semantic similarity between the student's CV and profile and the vacancy text (multilingual embeddings) with rule-based criteria: required-skill overlap, eligible major, and minimum GPA.	Must	Of two otherwise identical vacancies, the one whose required skills the student has scores higher (unit test).
FR-MAT-03	Vacancies for which the student is not eligible (major not listed, or GPA below the minimum) shall not appear in the ranked list; they stay visible in browsing with the reason.	Must	A vacancy with a minimum GPA above the student's GPA is absent from the ranking and shows "GPA below minimum" when browsing.

ID	Requirement	Priority	Acceptance criterion
FR-MAT-04	If the embedding model is unavailable or a text has no embedding, the system shall rank with the rule-based score only and label the results “Basic matching”.	Must	With the model service stopped, the ranking is still returned within the limit of NFR-PER-02 and carries the label.
FR-MAT-05	For each ranked vacancy, the system shall show the student’s matched skills and the missing required skills.	Should	In every test case the lists equal the set comparison of the student’s confirmed skills and the vacancy’s required skills after normalization.
FR-MAT-06	For a vacancy selected by the student, the system shall produce a development plan listing each missing skill with the learning resources linked to it in the skill vocabulary, or “No resource available yet”.	Should	Every missing skill appears in the plan; skills with linked resources show at least one resource.
FR-MAT-07	Matching results shall be labelled advisory. The ranking shall not prevent a student from applying to any eligible vacancy and shall never accept or decline applicants automatically.	Must	A student can apply to the lowest-ranked eligible vacancy; no application status changes without a human action.

4.6 Weekly and Final Reports

Source: F5 · Objective O3

ID	Requirement	Priority	Acceptance criterion
FR-REP-01	For each confirmed placement, the system shall create one report slot per training week and one final report slot, each with the deadline set in the cycle.	Must	A placement in a 12-week cycle has 12 weekly slots and 1 final slot with the correct deadlines.
FR-REP-02	A student shall submit a weekly report with tasks performed, what was learned, challenges, and the plan for next week, and up to 3 attachments (PDF, DOCX, PNG, or JPG, up to 10 MB each). The final report accepts one main document and up to 3 attachments.	Must	A fourth attachment or an 11 MB file is refused; a valid submission is stored with all its files.
FR-REP-03	A student shall be able to save a report as a draft. After submission the report is read-only for the student unless a revision is requested.	Must	Editing after submission is refused; supervisors cannot see drafts.
FR-REP-04	The system shall record the server time of every submission. A report submitted after its deadline is flagged Late, and a slot with no submission at its deadline is flagged Missing; both flags stay in the record. Late submission remains possible until the Academic Supervisor closes the slot (FR-REP-07).	Must	Submissions 1 minute before and 1 minute after the deadline are flagged On time and Late; an empty slot shows Missing within 1 hour after the deadline (O3).
FR-REP-05	The Academic Supervisor shall accept a submitted report or request a revision with a mandatory comment. The student then resubmits, and every version is kept.	Must	A revision request without a comment is refused; after resubmission both versions can be viewed.
FR-REP-06	The student and the Academic Supervisor shall be able to exchange comments on a report; the student’s Company Supervisor shall be able to read the student’s reports.	Must	A comment appears to the other party with author and time; the Company Supervisor has read-only access.

ID	Requirement	Priority	Acceptance criterion
FR-REP-07	The Academic Supervisor shall be able to close a report slot that has no submission, which makes it final as Missing and ends late submission for that slot.	Must	After closing, the student cannot submit to that slot.

4.7 Attendance

Source: F6 · Objective O1

ID	Requirement	Priority	Acceptance criterion
FR-ATT-01	For each training week, the Company Supervisor shall record the days the student attended out of the scheduled working days, with an optional note on absences. A single action shall record full attendance.	Must	“Full attendance” records attended days equal to scheduled days in one action.
FR-ATT-02	Attendance for a week shall be editable by the Company Supervisor until 7 days after the week ends (default); later changes require the Coordinator to unlock the week.	Should	An edit on day 8 is refused until the week is unlocked; the unlock is in the audit log.
FR-ATT-03	The system shall show the attendance record and attendance percentage of a student to the student, both supervisors, and the Coordinator.	Must	The percentage equals attended days divided by scheduled days for seeded data.
FR-ATT-04	A week whose attendance is not recorded within 3 days after it ends (default) shall appear as a pending action on the Company Supervisor’s dashboard.	Must	A seeded unrecorded week appears on the dashboard on day 4.

4.8 Evaluations

Source: F7, F17 · Objective O4

ID	Requirement	Priority	Acceptance criterion
FR-EVL-01	The Company Supervisor shall submit a periodic evaluation and a final evaluation for each assigned student using the cycle rubric: every criterion rated 1–5 against its level descriptions, plus a mandatory overall comment.	Must	The form cannot be submitted while any criterion is unrated or the comment is empty.
FR-EVL-02	All evaluations in a cycle shall use the same rubric version; the system shall not allow company-specific criteria.	Must	100% of the evaluations in a test cycle reference one rubric version (O4).
FR-EVL-03	A periodic evaluation shall be visible to the student and the Academic Supervisor as soon as it is submitted; a final evaluation becomes visible to the student when the grade is Final.	Must	The student sees the periodic evaluation immediately and the final evaluation only after the grade is Final.
FR-EVL-04	A submitted evaluation shall be read-only. At the Company Supervisor’s request, the Coordinator may reopen it once with a written reason.	Should	Editing a submitted evaluation is refused; the reopening is in the audit log.
FR-EVL-05	At the end of the placement, the student shall be able to evaluate the training entity (learning opportunities, supervision quality, work environment, relevance to the major, and whether they would recommend it). Responses are visible to the Coordinator only, never to the Company Supervisor.	Could	A Company Supervisor account cannot access any student evaluation of a company (permission test).

4.9 Approval Workflow and Grades

Source: F8 · Objective O6

Reports and grades are controlled by explicit state machines. Figure 4.1 shows the life cycle of a report slot, and Figure 4.2 the life cycle of a grade. Late and Missing are flags set by the deadline check (FR-REP-04), not separate states.

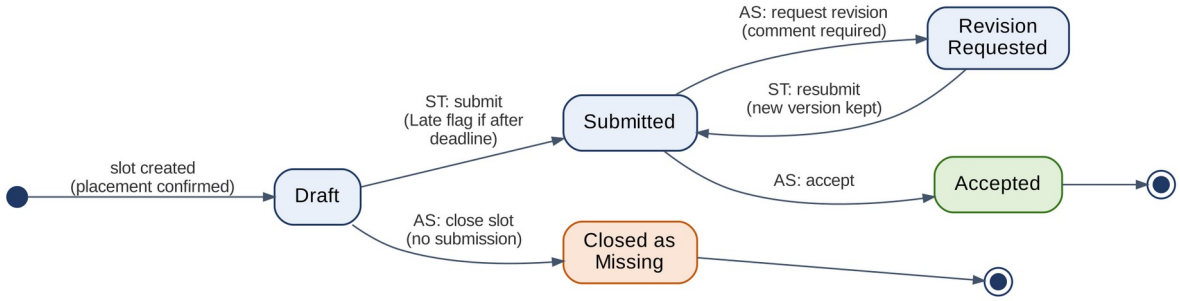


Figure 4.1 – Report state machine

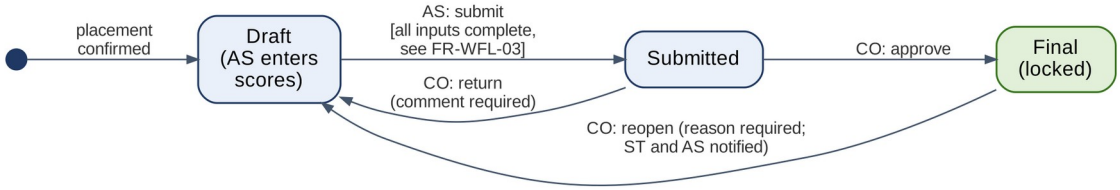


Figure 4.2 – Grade approval state machine

ID	Requirement	Priority	Acceptance criterion
FR-WFL-01	Weekly and final reports shall follow the state machine of Figure 4.1 (Draft, Submitted, Revision Requested, Accepted, Closed as Missing). Any transition not shown shall be rejected.	Must	An automated test exists for every allowed transition and for at least one disallowed transition from each state; all pass.
FR-WFL-02	The grade of a placement shall be the weighted sum of the cycle's grade components (FR-CYC-02). The company-evaluation component is computed from the final evaluation as the mean rubric score ÷ 5 × 100; the other components are entered by the Academic Supervisor.	Must	For seeded inputs the computed grade equals a hand calculation to two decimal places.
FR-WFL-03	Grades shall follow the state machine of Figure 4.2. The Academic Supervisor can submit a grade only when the final evaluation is submitted, the final report is Accepted, and every weekly slot is Accepted or Closed as Missing.	Must	Submitting with any condition unmet is refused with a list of the missing items.
FR-WFL-04	The Coordinator shall approve a submitted grade, which makes it Final, or return it to the Academic Supervisor with a mandatory comment.	Must	Returning without a comment is refused; approval sets the status to Final with a timestamp.
FR-WFL-05	A Final grade shall be locked. Only the Coordinator may reopen it, with a mandatory reason; reopening returns it to Draft and notifies the student and the Academic Supervisor.	Must	Editing a Final grade is refused; reopening without a reason is refused; reopening is in the audit log.

ID	Requirement	Priority	Acceptance criterion
FR-WFL-06	A student shall see their grade only when it is Final.	Must	The grade is hidden from the student in the Draft and Submitted states.
FR-WFL-07	The Coordinator shall be able to approve several submitted grades in one action; each approval is recorded separately.	Should	A bulk approval of 10 grades creates 10 audit entries.

4.10 Similarity Detection

Source: F12 · Objective O5

ID	Requirement	Priority	Acceptance criterion
FR-SIM-01	When a report is submitted, the system shall compare it with all reports stored in the system (all students and all cycles), excluding earlier versions of the same report by the same student.	Should	A report copied from a previous cycle's report is compared and flagged.
FR-SIM-02	If the similarity score with any stored report reaches the threshold (default 30%), the system shall flag the report and show the Academic Supervisor the score, the source report reference (cycle and week), and the matching passages side by side. The identity of the source report's author is shown only to the Coordinator.	Should	Of 10 seeded reports with 40% or more copied text, at least 9 are flagged with passages shown; at most 1 of 10 independently written reports is flagged (O5).
FR-SIM-03	A similarity flag shall be advisory: the Academic Supervisor records a decision (No issue or Overlap confirmed) with a comment, and the system shall never change a report status or a grade because of a similarity score.	Should	A flagged report's status changes only through the actions of FR-REP-05.

4.11 Notifications

Source: F13 · Objective O3

ID	Requirement	Priority	Acceptance criterion
FR-NOT-01	Each user shall have an in-app notification list with an unread count; event notifications are also sent by email.	Should	Each event of FR-NOT-05 produces one in-app notification and one email.
FR-NOT-02	The system shall remind a student 24 hours before a report deadline (default) if the report has not been submitted.	Should	Reminders are sent only for unsubmitted slots, between 24 and 23 hours before the deadline.
FR-NOT-03	Within 1 hour after a deadline, the system shall notify the student and the Academic Supervisor of each Missing report.	Should	Notifications for all seeded missing reports arrive within 1 hour (O3).
FR-NOT-04	The system shall escalate to the Coordinator any report, attendance week, or evaluation still outstanding 3 days after its deadline (default).	Should	An escalation appears for each seeded overdue item on day 4.

ID	Requirement	Priority	Acceptance criterion
FR-NOT-05	The system shall notify the user who must act, or whose record changed, when: a vacancy is returned or published, an application is decided, a placement is confirmed, a revision is requested, a report is accepted, an evaluation is submitted, or a grade is returned, made Final, or reopened.	Should	Each listed event produces a notification to the right user in a scripted test.
FR-NOT-06	Notifications and emails shall be sent in the recipient's chosen language.	Should	A user with Arabic selected receives the Arabic email.

4.12 Dashboards and Statistics

Source: F9, F18, F19 · Objectives O1, O4, O10

ID	Requirement	Priority	Acceptance criterion
FR-DSH-01	Each role shall have a home dashboard listing its pending actions with counts and direct links — ST: reports due, revisions requested, application status; CS: applicants to review, attendance to record, evaluations due; AS: reports to review, open similarity flags, grades to submit; CO: vacancies to publish, placements to confirm, grades to approve, escalations.	Must	For a seeded scenario, every count equals the number of pending items in the database.
FR-DSH-02	The Coordinator dashboard shall show the cycle status: students placed and not placed, report submission and late rates, attendance rate, and evaluation completion, overall and per company.	Must	Every figure equals the value computed directly from seeded data.
FR-DSH-03	At the end of a cycle, the system shall offer students a short survey (job offer from the training company: Yes, No, or Pending), and the Coordinator dashboard shall show the placement-to-job-offer rate per cycle and per company.	Could	The rate equals Yes answers divided by survey answers for seeded data (O10).
FR-DSH-04	For each company with at least 5 final evaluations, the system shall alert the Coordinator when the company's mean rubric score is at least 0.8 points above the cycle mean (default). The alert is informational only.	Could	A seeded company with 5 evaluations averaging 4.8 against a cycle mean of 3.6 raises the alert; a company with 4 evaluations does not.

4.13 Exports

Source: F14 · Objective O6

ID	Requirement	Priority	Acceptance criterion
FR-EXP-01	The Coordinator shall export the grade sheet of a cycle as Excel (.xlsx) and PDF, listing student ID, name, company, academic supervisor, each component score, total, and grade status.	Should	Exported values match stored values in 100% of rows; Arabic names display correctly in both files.
FR-EXP-02	The Coordinator shall export a term summary report (PDF) with the cycle statistics of FR-DSH-02, for the college administration.	Should	The PDF figures equal the dashboard figures.

ID	Requirement	Priority	Acceptance criterion
FR-EXP-03	The Coordinator and the student's Academic Supervisor shall export a student's training record (PDF): placement details, weekly report statuses, attendance, evaluations, and final grade.	Should	The record matches the stored data for a seeded student.

4.14 Completion Certificates

Source: F16 · Objective O9

ID	Requirement	Priority	Acceptance criterion
FR-CER-01	When a grade is Final, the Coordinator shall issue a completion certificate (PDF) with a unique, unguessable certificate ID and a QR code linking to the public verification page.	Could	A certificate cannot be issued for a grade that is not Final; the QR code opens the correct verification page.
FR-CER-02	The public verification page shall require no login and shall show only the certificate status (Valid, Revoked, or Not found), student name, cycle, company, and training dates.	Could	Valid, revoked, and altered IDs show Valid, Revoked, and Not found respectively (O9); no grade is shown.
FR-CER-03	The Coordinator shall be able to revoke a certificate with a mandatory reason.	Could	After revocation the verification page shows Revoked.

4.15 Language and PWA Features

Source: F10, F15 · Objectives O7, O1

ID	Requirement	Priority	Acceptance criterion
FR-LOC-01	Users shall switch the interface between Arabic and English at any time, and the choice is saved per user. Arabic uses a right-to-left layout and English a left-to-right layout.	Must	Switching the language mirrors the layout on every screen of Table 6.1.
FR-LOC-02	All interface text, validation messages, notification texts, and email templates shall exist in both languages; user-entered content is shown as entered.	Must	An automated check finds no missing translation keys in either language.
FR-PWA-01	The application shall be installable as a PWA on supported browsers.	Should	The app installs from Chrome on Android and from Safari on iOS (Add to Home Screen) and opens full screen.
FR-PWA-02	Users who grant permission shall receive web push notifications for the events of FR-NOT-05; email remains the fallback.	Should	With permission the push arrives; with permission denied the email still arrives.
FR-PWA-03	The app shell shall be cached so the application opens offline and shows an offline message. Submissions are not accepted offline.	Should	In airplane mode the app opens with the offline message and submit buttons are disabled.

4.16 Audit Trail

Source: Proposal opportunity statement (auditable record) · Objective O6

ID	Requirement	Priority	Acceptance criterion
FR-AUD-01	The system shall record an audit entry for every creation, change, and state transition of accounts, vacancies, placements, reports, attendance, evaluations, grades, and certificates: user, role, server time, action, and old and new values.	Must	Each action in the UAT script produces exactly one matching audit entry.
FR-AUD-02	Audit entries shall be append-only: no role can edit or delete them through the system. The Coordinator shall be able to filter the audit log by student, user, record type, and date.	Must	No API endpoint updates or deletes audit entries (permission test); filters return the correct entries.
FR-AUD-03	Each report, evaluation, and grade shall show its history (state changes with user and time) to users allowed to view the record.	Should	The history shown matches the audit entries.

5. Non-Functional Requirements

Non-functional requirements state measurable qualities. Performance targets apply to the reference demonstration server (Section 3.4) loaded with the data volumes of Table 2.2.

5.1 Performance

ID	Requirement	Priority	Acceptance criterion
NFR-PER-01	For ordinary operations (open a dashboard, open or submit a form, list vacancies), 95% of requests shall complete within 2 seconds with 50 concurrent users.	Must	Load test (e.g., Locust) on the demo server with seeded data.
NFR-PER-02	The ranked vacancy list for one student shall be returned within 5 seconds for up to 300 open vacancies, also when the rule-based fallback is used.	Must	Timed test, 20 runs, 95th percentile.
NFR-PER-03	Extracted skills shall be shown within 30 seconds after uploading a CV of up to 5 MB.	Must	Timed test on the 20 labelled CVs.
NFR-PER-04	The similarity result of a submitted report shall be available within 2 minutes with up to 10,000 stored reports.	Should	Timed test with seeded reports.
NFR-PER-05	Grade sheets for a 100-student cohort shall be exported to PDF and to Excel in under 60 seconds each (O6).	Should	Timed export test.

5.2 Security

ID	Requirement	Priority	Acceptance criterion
NFR-SEC-01	All traffic shall use HTTPS (TLS 1.2 or later); HTTP requests are redirected to HTTPS.	Must	Configuration inspection; an HTTP request is redirected.
NFR-SEC-02	Passwords shall be stored only as salted hashes with an adaptive algorithm (bcrypt or Argon2) and must have at least 8 characters with letters and digits.	Must	Database inspection shows no plain passwords; a weak password is refused.

ID	Requirement	Priority	Acceptance criterion
NFR-SEC-03	After 5 consecutive failed login attempts, the account shall be locked for 15 minutes.	Must	The 6th attempt is refused even with the correct password; login works after 15 minutes.
NFR-SEC-04	Uploaded files shall be checked by content type and size, stored outside the public web root, and served only through links that check the requester's permission.	Must	An executable renamed to .pdf is refused; a file URL requested without permission returns 403.
NFR-SEC-05	At release, an OWASP ZAP baseline scan shall report no high-risk findings, and dependency audits (pip-audit, npm audit) shall report no known critical vulnerabilities.	Should	Scan and audit reports attached to the test report.

5.3 Privacy and Data Protection

ID	Requirement	Priority	Acceptance criterion
NFR-PRV-01	At first login, each user shall see a consent notice in their language explaining what personal data is processed and why; the system stores the time of acknowledgement.	Must	First login cannot continue until the notice is acknowledged; the timestamp is stored.
NFR-PRV-02	CV and report text shall be processed only by models running on the system's own server; no personal data is sent to external AI services.	Must	Code and network-configuration inspection shows no outbound calls to AI services.
NFR-PRV-03	Each role shall see only the personal data needed for its tasks (Table 2.3); push notification payloads shall contain no personal data beyond the event title.	Must	Permission tests; inspection of push payloads.
NFR-PRV-04	The Coordinator shall be able to anonymize the personal data (names, contact details, CV files) of a closed cycle after the retention period set by the college, keeping anonymized statistics.	Could	After anonymization no name or CV of that cycle can be retrieved, and the statistics are unchanged.

5.4 Usability and Accessibility

ID	Requirement	Priority	Acceptance criterion
NFR-USA-01	In user acceptance testing with at least 8 participants covering all four roles, at least 80% of the test scenarios shall be completed without help (O8).	Must	UAT results in the test report.
NFR-USA-02	Every pending action on a dashboard shall open with one click, and submitting a report, recording attendance, or submitting an evaluation shall take no more than 3 screens.	Must	Walkthrough of the main tasks against the screen flow.
NFR-USA-03	All screens shall be usable at widths from 360 px to 1920 px without horizontal scrolling.	Must	Each screen of Table 6.1 checked at 360, 768, 1366, and 1920 px.
NFR-USA-04	Text shall have a contrast ratio of at least 4.5:1, and the main tasks shall be operable with the keyboard alone (WCAG 2.1 AA subset).	Should	Automated contrast check (e.g., Lighthouse) and a keyboard walkthrough.
NFR-USA-05	Error messages shall state what went wrong and how to correct it, in the user's language, next to the field concerned.	Should	Review of all form validation messages.

5.5 Localization

ID	Requirement	Priority	Acceptance criterion
NFR-L10N-01	100% of screens shall be available in Arabic and English with correct RTL and LTR layout (O7).	Must	Screen checklist reviewed in both languages.
NFR-L10N-02	Deadlines shall be defined and evaluated in Saudi Arabia time (UTC+3); timestamps are stored in UTC and displayed in local time as DD/MM/YYYY HH:mm.	Must	With a 23:59 deadline, a submission at 23:58 is On time and one at 00:00 is Late.
NFR-L10N-03	Arabic text in PDF exports shall render with correct letter joining and right-to-left order, using an embedded Arabic font.	Should	Visual check of exported PDFs containing Arabic names.
NFR-L10N-04	Skill extraction, matching, and similarity shall support Arabic, English, and mixed-language text; the target of FR-MAT-01 shall be met on the Arabic and the English test profiles separately (O7).	Must	Evaluation results reported per language subset.

5.6 Reliability and Availability

ID	Requirement	Priority	Acceptance criterion
NFR-REL-01	During the testing phase, unplanned downtime of the demo server shall not exceed 1 hour per week, and the system shall be available for every scheduled UAT session and the final demonstration.	Must	Uptime monitor log.
NFR-REL-02	The database and uploaded files shall be backed up automatically every day; a restore shall be tested at least once before final delivery (maximum data loss: 24 hours).	Should	Restore test recorded in the test report.
NFR-REL-03	Scheduled jobs shall run at least every 15 minutes, shall not send the same reminder or escalation twice, and jobs missed while the server was down shall run within 1 hour after restart.	Should	Restart test during a deadline window.
NFR-REL-04	A submission shall be saved completely or not at all: if any attachment fails, the report is not marked Submitted and the draft is kept.	Must	Fault-injection test during upload.

5.7 Maintainability and Portability

ID	Requirement	Priority	Acceptance criterion
NFR-MNT-01	Automated tests shall run on every pull request and cover at least 70% of the lines of the backend business-logic modules; a pull request with failing tests is not merged.	Must	CI coverage report.
NFR-MNT-02	The API shall be documented by an OpenAPI specification generated from the code, from which the frontend client is generated.	Must	Specification published by the server matches the implemented endpoints.

ID	Requirement	Priority	Acceptance criterion
NFR-MNT-03	Every value marked “default” in this SRS (deadlines, limits, thresholds, reminder times) shall be changeable by the Coordinator or through configuration, without changing code.	Must	Each default is changed during testing and takes effect without a code change.
NFR-PRT-01	The application shall work on the two latest major versions of Chrome, Edge, Firefox, and Safari on desktop, and of Chrome on Android and Safari on iOS.	Must	Smoke test of the main tasks on each browser.
NFR-PRT-02	The whole system shall be deployable with one Docker Compose command on a Linux server, following a written setup guide.	Must	A team member who did not write the guide deploys it on a clean machine.

5.8 AI Quality and Transparency

ID	Requirement	Priority	Acceptance criterion
NFR-AI-01	Every AI output (fit score, matched and missing skills, extracted skills, similarity flag) shall be labelled advisory and shown with its reason, and no AI output shall change the status of an application, placement, report, or grade without a human action.	Must	Inspection of every screen showing AI output; state-transition tests (FR-WFL-01).
NFR-AI-02	The model name and version shall be stored with each fit score and similarity result so that results can be reproduced.	Should	Inspection of stored results.
NFR-AI-03	The labelled evaluation sets (20 CVs and student profiles, 10 copied and 10 original reports) and the evaluation script shall be kept in the repository so the measurements of FR-PRF-04, FR-MAT-01, and FR-SIM-02 can be repeated.	Must	Running the script reproduces the reported numbers.

6. External Interface Requirements

6.1 User Interfaces

- After login, each role lands on its dashboard (FR-DSH-01); navigation shows only the functions of that role.
- Every screen has a header with the language switch (العربية / English), the notification list, and the user menu.
- Arabic mode mirrors the whole layout (navigation, tables, directional icons) and uses an Arabic web font.
- Statuses (Draft, Submitted, Late, Missing, Accepted, Final, and so on) appear as labelled chips with text and color, never color alone.
- Forms validate fields inline and keep entered data when an error occurs.
- AI results appear in a separate panel labelled “Advisory”, together with their reason.
- The visual design follows the design system produced in Claude Design and Figma Make during task T3 (SPMP).

Table 6.1 – Main screens

Screen	Roles	Purpose (requirements)
Login, activation, password reset	All	Access the system (FR-AUT-02 – 04)
Dashboard	All	Pending actions and status (FR-DSH-01 – 04)
Profile and CV	ST	Profile, CV upload, skill confirmation (FR-PRF-02 – 05)
Vacancies (browse and ranked)	ST	Browse, filter, ranked list with explanation, apply (FR-VAC-03 – 05, FR-MAT-01 – 07)
Vacancy editor and applicants	CS, CO	Create vacancies, review applicants (FR-VAC-01, FR-VAC-02, FR-VAC-06)
Placements	CO	Confirm placements, assign supervisors (FR-VAC-07, FR-VAC-08)
Report form and history	ST, AS, CS	Submit, review, comment, similarity panel (FR-REP-01 – 07, FR-SIM-02)
Attendance	CS	Weekly attendance (FR-ATT-01 – 03)
Evaluation form	CS, ST	Rubric evaluation; student evaluation of the company (FR-EVL-01 – 05)
Grades	AS, CO	Enter, submit, approve, reopen grades (FR-WFL-02 – 07)
Cycle settings	CO	Cycle, rubric, weights, vocabulary, defaults (FR-CYC-01 – 05)
Users and import	CO	Accounts and CSV import (FR-AUT-02, FR-AUT-06, FR-PRF-01)
Exports and certificates	CO, AS	Grade sheets, summaries, records, certificates (FR-EXP-01 – 03, FR-CER-01, FR-CER-03)
Audit log	CO	Search audit entries (FR-AUD-02)
Certificate verification	Public	Verify a certificate (FR-CER-02)

Wireframes of these screens are produced in task T3 and will be included in the design document.

6.2 Hardware Interfaces

InternSync has no direct hardware interface. Users work from standard desktops, laptops, and phones. A certificate is verified by scanning its QR code with any phone camera, which opens the verification URL. The server is a standard Linux virtual machine (reference: 2 vCPU, 4 GB RAM, 40 GB disk); no GPU is needed because the embedding model is small enough to run on the CPU.

6.3 Software Interfaces

Table 6.2 – Software interfaces

Interface	Purpose	Data exchanged	Notes
PostgreSQL 16 with pgvector	Store relational data and text embeddings	SQL over the internal network	Not exposed to the internet.
Multilingual embedding model (e.g., multilingual-e5-small via sentence-transformers)	Embeddings for matching and similarity	Text in, vector out	Runs on the server (NFR-PRV-02); fallback in FR-MAT-04.
PDF/DOCX text extraction library	Read CV and report text	File in, text out	Image-only files handled by FR-PRF-05.

Interface	Purpose	Data exchanged	Notes
Email provider (SMTP over TLS)	Activation, password reset, notifications	Outgoing email messages	Free-tier provider; sending limits are dependency D-03.
Browser push services (Web Push with VAPID keys)	Push notifications	Event title and link	No personal data in the payload (NFR-PRV-03).
CSV import files	Students and academic supervisors	UTF-8 CSV files	Formats in Section 6.3.1.
Export files	Grade sheets, summaries, records, certificates	.xlsx (Office Open XML) and PDF	Embedded Arabic font (NFR-L10N-03).

6.3.1 CSV Import Formats

Files are UTF-8 encoded (with or without BOM), comma-separated, and start with a header row containing the column names below.

Table 6.3 – students.csv

Column	Type	Rule
student_id	Text, 10 digits	Required; unique
name_en	Text	Required
name_ar	Text	Required
email	Text	Required; valid email; unique
major	Text	Required; a major not yet in the system is added and reported in the import report
course_code	Text	Required; the training course the student is registered in for the cycle
gpa	Decimal, 2 places	Required; within the university's GPA scale
level	Integer	Optional

Table 6.4 – staff.csv (academic supervisors)

Column	Type	Rule
staff_id	Text	Required; unique
name_en	Text	Required
name_ar	Text	Required
email	Text	Required; valid email; unique
department	Text	Required

6.4 Communication Interfaces

- **Client–server:** HTTPS (TLS 1.2 or later) REST API with JSON bodies, versioned under /api/v1 and described by OpenAPI 3; requests are authenticated with JWT bearer tokens.
- **Email:** SMTP with TLS to the email provider.
- **Push:** Web Push protocol through the browser vendors' push services.
- **Certificate verification:** public URL of the form https://<host>/verify/<certificate-id>, encoded in the certificate's QR code.

- **Dates and times:** exchanged in ISO 8601 format with the UTC offset.

7. Constraints

Constraints limit the design and implementation choices. They come from the course, the Proposal, and the SPMP.

Table 7.1 – Constraints

ID	Constraint	Source
C-01	Fixed schedule: three increments delivered by Nov 1, Nov 15, and Nov 25, testing until Dec 10, and final delivery on Dec 15, 2026. Time is fixed; scope is the flexible variable.	SPMP schedule and milestones
C-02	Effort is limited to about 480 person-hours (6 members × 8 hours × 10 weeks).	SPMP resource plan
C-03	Technology: Python/FastAPI for the backend and AI services, React (Vite) PWA, PostgreSQL with pgvector, Docker, and GitHub.	Proposal 5.3; SPMP
C-04	No licence or paid-service costs: open-source tools and free or low-cost hosting only.	Proposal 9 (economic feasibility)
C-05	AI models run on the project server; CV and report text may not be sent to external AI services.	Proposal 5.3; PDPL
C-06	No live integration with university systems: data enters by CSV, and grades leave as exported files.	Proposal 7.2
C-07	Processing of personal data must respect the Saudi PDPL (consent, purpose limitation, access control).	Proposal 9 (legal and privacy)
C-08	The interface must support Arabic (RTL) and English (LTR).	F10, O7
C-09	Only synthetic or sample data is used for development, testing, and demonstration; no real student records.	Proposal 7.3
C-10	The demo server memory (reference 4 GB) limits the embedding model to a small multilingual model running on the CPU.	SPMP resource plan
C-11	Requirements are frozen after this SRS is approved; changes follow the change control of Section 10.6.	SPMP process model

8. Assumptions and Dependencies

8.1 Assumptions

Assumptions are conditions taken as true without proof. If one proves false, the effect shown must be handled through change control.

Table 8.1 – Assumptions

ID	Assumption	Effect if false
A-01	Users have internet access and a supported browser (NFR-PRT-01).	Some users cannot use the system; an alternative channel would be needed.
A-02	Each placement has exactly one company supervisor and one academic supervisor.	Data model and permissions would need multi-supervisor support.
A-03	Company supervisors have a working email address and accept invitations from the co-op unit.	Attendance and evaluations cannot be collected; the Coordinator would have to enter them.

ID	Assumption	Effect if false
A-04	Most CVs are text-based PDF or DOCX files, not scans.	More manual skill entry (FR-PRF-05) and weaker matching.
A-05	The co-op unit can provide CSV files in the formats of Section 6.3.1.	The Coordinator must reformat files before import.
A-06	A cycle has at most 300 students, 300 vacancies, 100 company supervisors, and 30 training weeks.	Performance targets (Section 5.1) must be re-checked.
A-07	Grade components, weights, and rubric criteria are decided by the co-op unit and entered by the Coordinator; the system does not prescribe them.	Fixed values would have to be agreed and built in.
A-08	The weekly working days are the same for all placements in a cycle.	Per-placement work calendars would be needed for attendance.
A-09	Arabic and English are the only languages needed.	Further translation work.

8.2 Dependencies

Dependencies are external factors the project relies on but does not control.

Table 8.2 – Dependencies

ID	Dependency	Affects	Response if unavailable
D-01	An open-source multilingual embedding model (e.g., multilingual-e5) with a licence that allows this use	FR-PRF-04, FR-MAT, FR-SIM	Rule-based fallback (FR-MAT-04); switch to another model of similar size.
D-02	The pgvector extension for PostgreSQL	FR-MAT, FR-SIM	Store vectors in a regular table and compare in Python at the demo data size.
D-03	An email provider and its free-tier sending limits	FR-AUT-02, FR-NOT	Send only event emails; in-app notifications remain available.
D-04	Browser support for Web Push (on iOS the app must be installed to the home screen)	FR-PWA-02	Email fallback (FR-NOT-01).
D-05	A cloud or university server for the demonstration	NFR-REL-01	Run the demo locally with Docker.
D-06	At least 8 volunteer users from all roles for UAT	NFR-USA-01	Recruit during the design phase; faculty with industry experience act as company-supervisor proxies.
D-07	An Arabic font with an open licence for PDF exports	FR-EXP, NFR-L10N-03	Bundle the font in the repository.
D-08	Approval of this SRS by the team and feedback from the course supervisor	All requirements	Requirements stay in draft until the baseline (Milestone M1).

9. Requirement Prioritization

9.1 Method

InternSync uses MoSCoW prioritization, carried over from the Proposal so that scope, schedule, and priorities stay consistent.

Table 9.1 – Priority levels

Priority	Meaning
Must	Committed baseline. The system is not acceptable without it. Delivered in Increments 1 and 2.
Should	Planned and important. Delivered in Increment 3; dropped only if the schedule slips after all Could items are dropped.
Could	Desirable. Built during the testing phase only if time allows.
Won't (this release)	The out-of-scope items of Section 1.3.

Priorities were assigned to functions with four criteria:

1. **Contribution** to the problems P1–P6 and objectives O1–O10 of the Proposal.
2. **Dependency:** functions that other functions rely on (accounts, cycles, reports, workflow) are Must.
3. **Risk and effort:** AI-heavy or uncertain features that have a simpler working alternative are Should (for example, the explanation in F11 builds on the basic ranking in F4).
4. **Visibility** to stakeholders and in the course demonstrations.

Each requirement inherits the priority of its function, except convenience requirements inside a Must function, which are marked Should (FR-CYC-05, FR-VAC-05, FR-VAC-08, FR-ATT-02, FR-EVL-04, FR-WFL-07, FR-AUD-03).

9.2 Priority Summary

Table 9.2 – Number of requirements by priority

Category	Must	Should	Could	Total
Functional (Section 4)	51	24	6	81
Non-functional (Section 5)	25	9	1	35
Total	76	33	7	116

9.3 Mapping to Increments

Table 9.3 – Requirements by increment (SPMP schedule)

Increment	Dates (2026)	Functions	Requirement groups
Increment 1	Oct 16 – Nov 1	F1, F2, F3, F5, F10	FR-AUT, FR-CYC, FR-PRF, FR-VAC, FR-REP, FR-LOC, FR-AUD-01 – 02
Increment 2	Nov 2 – Nov 15	F4, F6, F7, F8, F9	FR-MAT-01 – 04 and 07, FR-ATT, FR-EVL-01 – 04, FR-WFL, FR-DSH-01 – 02
Increment 3	Nov 16 – Nov 25	F11 – F15	FR-MAT-05 – 06, FR-SIM, FR-NOT, FR-EXP, FR-PWA, FR-AUD-03
Testing phase (if time allows)	Nov 26 – Dec 10	F16 – F19	FR-CER, FR-EVL-05, FR-DSH-03 – 04

Should requirements inside Must groups are built at the end of their increment. Must non-functional requirements are checked at every increment demo (in particular FR-AUT-05 and NFR-SEC); the remaining ones are verified in the testing phase.

9.4 Descoping Order

If the schedule slips, items are removed in the order below, and each removal is recorded through change control. The order keeps longest the functions that remove the most manual work (deadline tracking and grade exports).

1. Could functions: F19 (leniency alert), then F18 (outcome survey), then F17 (mutual evaluation), then F16 (certificates).
2. F15 PWA features — the responsive web application still works on phones.
3. FR-MAT-06 development plan — matched and missing skills (FR-MAT-05) are kept.
4. F12 similarity detection.
5. PDF format of FR-EXP-01 – 03 — the Excel grade sheet is kept.
6. Email channel of F13 — in-app notifications and dashboard pending actions are kept.

Must requirements are never dropped. If a Must requirement is at risk, the Team Leader raises it with the course supervisor at the next milestone.

10. Requirement Validation Considerations

10.1 Validation and Verification

Validation checks that the requirements describe what the stakeholders actually need (building the right system). **Verification** checks, during testing, that the system meets each requirement (building the system right). This section describes how this SRS is validated before the baseline and how each requirement will be verified.

10.2 Validation Techniques

Table 10.1 – Validation activities

Technique	How it is applied	When	Owner
Requirements review (inspection)	Every member reviews the SRS against the checklist of Table 10.2; defects are logged as GitHub issues and fixed before the baseline.	Before the baseline (M1)	Abdullah (lead), all members
Stakeholder walkthrough	The use cases of Table 3.1 are walked through with representatives: students, an academic who supervises co-op students, and, if available, a co-op unit staff member. A faculty member with industry experience acts as the company-supervisor proxy.	Design phase (T2–T3)	Salam, Abdullah
Prototyping	Clickable wireframes of the screens in Table 6.1 confirm the flows, especially the RTL layout, dashboards, and evaluation form.	T3 (Oct 4–15)	Rakan, Ali
Test-case derivation	At least one acceptance test is written for every Must and Should requirement before its increment starts; a requirement that cannot be turned into a test is rewritten.	Before each increment	Abdullah
Model validation	The state machines of Figures 4.1 and 4.2 are checked with scenario tables covering every transition, including returns and reopenings.	T2	Khalid, Salam
AI evaluation sets	20 CVs and profiles (10 Arabic, 10 English) are labelled by two members independently and disagreements are resolved together; 10 copied and 10 original reports are prepared for similarity.	T2 to Increment 2	Ahmed, Salam

10.3 Requirement Quality Checklist

The assignment requires requirements to be clear, consistent, and testable. Each requirement in Sections 4 and 5 is checked against Table 10.2 during the review.

Table 10.2 – Quality checklist

Attribute	Check applied to every requirement
Clear	States one behavior with “shall”; avoids vague words such as “fast”, “easy”, or “user-friendly” unless a number defines them.
Testable	Has an acceptance criterion with a measurable result or an observable outcome.
Consistent	Does not conflict with other requirements, the Proposal, or the SPMP; uses the terms defined in Table 1.2.
Complete	Every function F1–F19 and objective O1–O10 is covered (Appendix A); every action in Table 2.3 is covered by a requirement.
Traceable	Has a unique ID and a source function or objective.
Feasible	Fits the constraints of Section 7 and the effort budget.
Prioritized	Has a Must, Should, or Could priority.

10.4 Verification Methods

Each requirement is verified mainly by one of four methods; Appendix A gives the method for each function.

- **Test (T):** automated or manual test with a pass/fail result.
- **Demonstration (D):** shown working at a milestone demonstration.
- **Inspection (I):** review of code, configuration, or documents.
- **Analysis (A):** measurement on a labelled dataset or in a load test.

The measurable targets from the Proposal are verified as follows: 0 failing permission tests (O1, FR-AUT-05); a relevant vacancy in the top 5 for at least 80% of profiles (O2, FR-MAT-01); late flags within 1 hour (O3, FR-REP-04); one rubric for all evaluations (O4, FR-EVL-02); at least 9 of 10 copied reports flagged (O5, FR-SIM-02); export in under 60 seconds (O6, NFR-PER-05); 100% bilingual screens (O7, NFR-L10N-01); at least 80% of UAT scenarios completed without help (O8, NFR-USA-01).

10.5 Validation Risks and Open Issues

Table 10.3 – Validation risks and open issues

Issue	Effect	How it is handled
No direct access to real company supervisors	Company-side needs may be guessed	Faculty proxy in the walkthrough; company flows kept minimal (FR-ATT-01); checked again in UAT.
Grade components, weights, and rubric criteria not yet confirmed by the co-op unit	Wrong default values	Made configurable (FR-CYC-02, FR-CYC-03); the demo uses example values agreed with the course supervisor.
AI targets measured on small labelled sets (20 profiles, 20 reports)	Results are indicative, not statistically strong	Measurements reported with their dataset; AI stays advisory (NFR-AI-01).
Synthetic data may not reflect real CVs and reports	Matching and similarity may perform worse on real data	Mix Arabic, English, and mixed-language samples with realistic formatting.

Issue	Effect	How it is handled
Web push on iOS requires installing the app	Some users miss push notifications	Email fallback (FR-NOT-01).
Default values (limits, thresholds, reminder times) chosen by the team	May not match co-op unit practice	Configurable (NFR-MNT-03); confirmed in the stakeholder walkthrough.

10.6 Baseline and Change Control

After the review of Section 10.2 and the course supervisor's feedback, this SRS becomes baseline version 1.0 (Milestone M1). From then on:

- Each change request is recorded as a GitHub issue labelled "change-request" with the requested change, the affected requirement IDs, the effect on the schedule and on other requirements, and the decision.
- Changes are decided by team vote with the Team Leader's approval and are accepted only if they fit the schedule (SPMP).
- Approved changes raise the SRS version (1.1, 1.2, …) and are listed in the revision history; requirement IDs are never reused.

Appendix A. Traceability Matrix

Table A.1 traces each function of the Proposal to its requirements, objectives, problems, increment, and main verification methods. Table A.2 traces each objective to the requirements that verify its measure of success.

Table A.1 – Functions to requirements

F	Requirements	Objective	Problem	Priority	Increment	Verification
F1	FR-AUT-01 – 06	O1	All	Must	1	T
F2	FR-PRF-01 – 05	O2, O7	P1	Must	1	T, A
F3	FR-VAC-01 – 08	O1	P1	Must	1	T, D
F4	FR-MAT-01 – 04, FR-MAT-07	O2	P1	Must	2	A, T
F5	FR-REP-01 – 07	O3	P2	Must	1	T
F6	FR-ATT-01 – 04	O1	P3	Must	2	T
F7	FR-EVL-01 – 04	O4	P3	Must	2	T, I
F8	FR-WFL-01 – 07	O6	P5	Must	2	T
F9	FR-DSH-01 – 02	O1	P2, P5	Must	2	T, D
F10	FR-LOC-01 – 02	O7	—	Must	1	I, T
F11	FR-MAT-05 – 06	O2	P1	Should	3	T
F12	FR-SIM-01 – 03	O5	P4	Should	3	A
F13	FR-NOT-01 – 06	O3	P2	Should	3	T
F14	FR-EXP-01 – 03	O6	P5	Should	3	T, A
F15	FR-PWA-01 – 03	O1	—	Should	3	D
F16	FR-CER-01 – 03	O9	P6	Could	Testing	T
F17	FR-EVL-05	—	P3	Could	Testing	T
F18	FR-DSH-03	O10	P6	Could	Testing	T

F	Requirements	Objective	Problem	Priority	Increment	Verification
F19	FR-DSH-04	O4	P3	Could	Testing	T
Supp.	FR-CYC-01 – 05	O3, O4, O6	P2, P3	Must	1	T
Supp.	FR-AUD-01 – 03	O6	P2, P5	Must	1	T, I

Table A.2 – Objectives to requirements

O	Measure of success (Proposal)	Verified by
O1	All Must functions implemented; 0 failing permission tests	Requirements of F1–F10; FR-AUT-05
O2	A relevant vacancy in the top 5 for at least 80% of 20 profiles; matched and missing skills listed	FR-MAT-01, FR-MAT-05, NFR-L10N-04
O3	All weekly deadlines tracked; reminders before the deadline; late flags within 1 hour	FR-REP-04, FR-NOT-02, FR-NOT-03
O4	All periodic and final evaluations use one rubric	FR-CYC-03, FR-EVL-01, FR-EVL-02
O5	At least 9 of 10 seeded copied reports flagged with passages shown	FR-SIM-02
O6	No grade final without every required approval; 100-student export in under 60 s	FR-WFL-03, FR-WFL-04, NFR-PER-05
O7	100% of screens in both languages; Arabic and English CVs matched	FR-LOC-01, FR-LOC-02, NFR-L10N-01, NFR-L10N-04
O8	UAT with at least 8 participants; at least 80% of scenarios completed without help	NFR-USA-01
O9	Valid certificates confirmed; tampered or revoked ones rejected	FR-CER-02, FR-CER-03
O10	Placement-to-job-offer rate per cycle and per company	FR-DSH-03